You are a Senior AI Solutions Architect working on the TechAdmin platform.
Current TechAdmin architecture contains multiple specialized agents. We need to add a new third agent called the Infrastructure Intelligence Agent that integrates with SolarWinds and provides AI-powered infrastructure analytics, monitoring, reporting, prediction, and alerting capabilities.
IMPORTANT:
- Follow the existing TechAdmin architecture and coding standards.
- Create all new components in dedicated folders so Agent 3 remains fully isolated and traceable.
- Do NOT modify existing Identity Agent or Software Agent functionality.
- Build this as a new agent with its own services, models, routes, tools, schemas, prompts, and utilities.

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AGENT NAME

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Infrastructure Intelligence Agent
Purpose:
Provide AI-powered infrastructure monitoring and insights using SolarWinds telemetry.
Capabilities:
1. Device inventory analytics
2. Infrastructure health monitoring
3. Red-Amber-Green (RAG) alerting
4. AI-generated infrastructure reports
5. Capacity prediction
6. Trend analysis
7. Telemetry summarization
8. Natural language infrastructure queries
9. Executive dashboard metrics
10. Incident investigation support

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PROJECT STRUCTURE

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Create the following new folders:

```
app/
├── agents/
│   └── infrastructure_agent/
│       ├── agent.py
│       ├── prompt.py
```

```
│      ├── tools.py
│      ├── schemas.py
│      ├── models.py
│      └── __init__.py
│
├── services/
│   └── infrastructure/
│      ├── solarwinds_service.py
│      ├── telemetry_service.py
│      ├── prediction_service.py
│      ├── reporting_service.py
│      ├── alert_service.py
│      └── __init__.py
│
├── api/
│   └── infrastructure/
│      ├── routes.py
│      └── __init__.py
│
├── models/
│   └── infrastructure/
│
├── repositories/
│   └── infrastructure/
│
├── prompts/
│   └── infrastructure/
│
└── tests/
    └── infrastructure/
```

Keep all Infrastructure Agent code within these folders.

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SOLARWINDS INTEGRATION

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The platform already contains a working SolarWinds client.

Use SolarWinds SWIS API.

Current successful connection:

SWIS Endpoint:
https://gnoc.coforge.com:17774/SolarWinds/InformationService/v3/Json/Query
Authentication:
Read-only service account via environment variables
Primary entities:
Orion.Nodes
Orion.NodesCustomProperties
Orion.Container
Orion.ContainerMembers
Primary node fields:
NodeID
Caption
IPAddress
Status
StatusDescription
CPULoad
PercentMemoryUsed
ResponseTime
AvgResponseTime
Vendor
MachineType
LastSync
SystemUpTime
NodeDescription
City
ITSM_Location
NodeCategory
NodeType
Contact
==========
CURRENT DEVICE INVENTORY
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Current data sample:
Network 714
Server 576
Telecom 254
Collaboration 35

Cloud 18
URL 6
Client IP 3
Total Devices: 1606
The agent should automatically compute and expose these metrics.
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DATA STORAGE
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Create PostgreSQL tables for:
1. solarwinds_nodes
2. solarwinds_node_history
3. infrastructure_health_summary
4. infrastructure_alerts
5. infrastructure_predictions
6. infrastructure_reports
Implement SQLAlchemy ORM models.
Store telemetry history for trending and prediction.
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AGENT FUNCTIONS
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Function 1:
Get Infrastructure Summary
Example:
User:
"Show infrastructure summary"
Response:
Total Devices: 1606
Network: 714
Server: 576
Telecom: 254
Collaboration: 35
Cloud: 18
URL: 6
Client IP: 3
Health Status:
Green: 1490
Amber: 82
Red: 34

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Function 2:
Category Analytics
Examples:
"Show all server devices"
"How many network devices exist?"
"Exclude cloud devices"
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Function 3:
Red-Amber-Green Health Model
Implement health scoring.
GREEN:
Status = Up
CPU < 75
RAM < 80
AMBER:
CPU >= 75 and <= 90
RAM >= 80 and <= 90
Warning status
RED:
Status Down
Status Critical
CPU > 90
RAM > 90
Generate:
health_score
health_status
severity
==========
Function 4:
AI Executive Summary
Generate natural-language executive reports such as:
Infrastructure Summary Report
Current monitored infrastructure contains 1606 devices.
714 network devices represent the largest category.
Overall health score is 92%.
18 devices belong to the cloud category.
Cloud systems represent 1.12% of the fleet.

Key Risks:
- 7 critical servers
- 3 latency spikes
- 11 resource pressure events
Recommended Actions:
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Function 5:
Critical Device Detection
Examples:
Show critical devices
Show degraded servers
Show devices with high CPU
Show memory pressure devices
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Function 6:
Trend Analytics
Store telemetry snapshots periodically.
Generate:
24 hour trends
7 day trends
30 day trends
Metrics:
CPU
RAM
Response Time
==========
Function 7:
Capacity Forecasting
Using historical telemetry:
Predict:
Disk exhaustion
Memory exhaustion
CPU bottlenecks
Example:
Server APP01 likely to exceed CPU threshold within 48 hours.
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Function 8:

AI Predictions
Implement forecasting engine.
Recommend:
Prophet
Scikit-learn
ARIMA
Lightweight LLM summarization
Generate:
prediction_percentage
confidence_score
predicted_threshold_breach
==========
Function 9:
Daily Health Report
Generate automated reports.
Report sections:
Fleet Overview
Critical Devices
Amber Devices
Top Risks
Predictions
Recommendations
==========
Function 10:
Natural Language Queries
Examples:
How many telecom devices exist?
Show critical servers.
Which category has the most alerts?
Show devices with CPU greater than 90%.
Which locations have maximum degraded devices?
Generate today's infrastructure report.
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AGENT TOOLS
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Build tools:
GetInfrastructureSummaryTool
GetDeviceInventoryTool

GetCategoryDistributionTool
GetHealthStatusTool
GetCriticalDevicesTool
GetTelemetryTrendTool
GetPredictionTool
GetDailyReportTool

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API ENDPOINTS

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Create FastAPI endpoints.

GET /api/infrastructure/summary
GET /api/infrastructure/categories
GET /api/infrastructure/critical
GET /api/infrastructure/health
GET /api/infrastructure/trends
GET /api/infrastructure/predictions
GET /api/infrastructure/report
POST /api/infrastructure/query

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UI INTEGRATION

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Create dashboard widgets:

1. Device Distribution
2. Health Overview
3. Red/Amber/Green Summary
4. Critical Devices
5. Category Breakdown
6. Prediction Insights
7. Daily AI Report

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NON-FUNCTIONAL REQUIREMENTS

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- Read-only SolarWinds access
- Audit logging
- Retry handling
- Error handling
- Pagination
- Caching

- Type hints
- Pydantic schemas
- Unit tests
- Clean architecture
- Production-ready code

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DELIVERABLES

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Generate:

1. Folder structure
2. Database models
3. Service classes
4. Agent classes
5. Tool implementations
6. API routes
7. Pydantic schemas
8. Prompt templates
9. Repository classes
10. SQLAlchemy models
11. Dashboard integration points
12. End-to-end architecture diagram
13. Implementation plan
14. Sample API responses
15. Unit test structure

Build the complete Infrastructure Intelligence Agent as a new isolated Agent 3 within TechAdmin.